TECHNICAL DOCUMENT

Service Dependency Map

docs/SERVICE_DEPENDENCY_MAP.md

Runtime · Build-time · Kafka · External APIs

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EduTech AI Platform — Service Dependency Map

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1. Dependency Matrix

The table below summarises what each service depends on. "X" indicates a hard runtime dependency; "proxy" means the gateway routes HTTP to that service without a direct application-layer dependency; "(P2)" indicates a Phase 2 planned dependency not yet active.

SERVICE	API-GATEWAY	AUTH-SVC	CENTER-SVC	PARENT-SVC	ASSESS-SVC	PSYCH-SVC	AI-GATEWAY-SVC	PSYCH-AI-SVC	ANTHROPIC API	OPENAI API	OLLAMA	HCAPTCHA API	SMTP
api-gateway	—	proxy	proxy	proxy	proxy	proxy	proxy						
auth-svc		—										X	X
center-svc			—				X						
parent-svc				—			X						
assess-svc					—		X						
psych-svc						—	X	X					
ai-gateway-svc							—	X	X	X	X(P2)		

Kafka topic flows (separate from HTTP):

TOPIC	PRODUCER	CONSUMER(S)
`auth-events`	auth-svc	(future: notification-svc)
`center-events`	center-svc	parent-svc, assess-svc
`parent-events`	parent-svc	(future: notification-svc)
`assess-events`	assess-svc	psych-svc
`psych-events`	psych-svc	(future: analytics-svc, notify-svc)
`ai-gateway-events`	ai-gateway-svc	(future: cost-analytics-svc)
`audit-immutable`	all services	(future: audit-svc — read-only)
`notification-send`	auth-svc	(future: notification-svc)

2. Per-Service Dependencies

2.1 api-gateway

Purpose: Edge proxy. Routes all inbound client traffic to downstream services. Enforces JWT RS256 authentication via a `GlobalFilter` and applies Redis token-bucket rate limiting to all requests.

Inbound: External clients (web browsers, mobile apps, API consumers).

Outbound HTTP (proxy routes):

ROUTE ID	PATH PREDICATE	DESTINATION ENV VAR
`auth-svc`	`/api/v1/auth/**`	`${AUTH_SVC_URI}`
`center-svc`	`/api/v1/centers/**`	`${CENTER_SVC_URI}`
`parent-svc`	`/api/v1/parents/**`	`${PARENT_SVC_URI}`
`assess-svc`	`/api/v1/assessments/**`	`${ASSESS_SVC_URI}`
`psych-svc`	`/api/v1/psych/**`	`${PSYCH_SVC_URI}`
`ai-gateway-svc`	`/api/v1/ai/**`	`${AI_GATEWAY_SVC_URI}`

Public paths that bypass JWT validation: `/api/v1/auth/**`, `/actuator/**`, `/swagger-ui/**`, `/v3/api-docs/**`, `/webjars/**`

Headers injected on valid JWT and forwarded to all downstream services: `X-User-Id`, `X-User-Role`, `X-User-Center-Id`, `X-Request-ID`

Outbound async (Kafka): None. The gateway publishes no Kafka events.

Database: None. The gateway is fully stateless — no JPA, no datasource configured.

Cache (Redis):
- Spring Cloud Gateway `RequestRateLimiter` default filter (Redis token bucket).
- Configuration: `redis-rate-limiter.replenish-rate`, `redis-rate-limiter.burst-capacity`, `redis-rate-limiter.requested-tokens` — all via env vars.
- Behaviour on Redis outage: rate limiting **fails open** — requests pass through unmetered. By design.

Resilience4j circuit breaker (auth-svc route): `sliding-window-size: ${R4J_CB_AUTH_WINDOW_SIZE}`, `failure-rate-threshold: ${R4J_CB_AUTH_FAILURE_THRESHOLD}`, `wait-duration-in-open-state: ${R4J_CB_AUTH_WAIT_DURATION}`

Runtime: Spring Cloud Gateway (WebFlux-based reactive runtime). `GlobalFilter` order `-2` injects `X-Request-ID`; order `-1` enforces JWT.

2.2 auth-svc

Purpose: Authentication and identity boundary. The **only** service permitted to issue, rotate, and revoke JWT access tokens. All other services validate tokens independently but never produce them.

Inbound:
- `api-gateway` (proxies `/api/v1/auth/**`)
- No direct calls from other services (JWT validation is done locally in each service, not via auth-svc at runtime)

Outbound HTTP (synchronous):

TARGET	ADAPTER CLASS	CALL	CIRCUIT BREAKER
hCaptcha Enterprise API	`HCaptchaRestAdapter` (Spring `RestClient`)	`POST ${HCAPTCHA_VERIFY_URL}` with `secret`, `response` (captcha token), `remoteip` as form body	`captcha-client` — fallback: `return false` (fail closed — reject on outage)

TARGET	ADAPTER CLASS	CALL	CIRCUIT BREAKER
SMTP server	`SmtpNotificationAdapter` (JavaMailSender)	Sends OTP emails for `EMAIL_VERIFICATION`, `PASSWORD_RESET`, `LOGIN_MFA` purposes	None — exception propagates as HTTP 500
Twilio Verify API	`SmtpNotificationAdapter.sendOtpSms()`	SMS OTP delivery (production; dev logs OTP to console)	None

Outbound async (Kafka topics published):

TOPIC (RESOLVED FROM ENV VAR)	EVENTS PUBLISHED	TRIGGER
`${KAFKA_TOPIC_AUTH_EVENTS}`	User registered, authenticated, logged out, token refreshed, OTP requested	On each auth use case completion
`${KAFKA_TOPIC_AUDIT_IMMUTABLE}`	All auth events duplicated to the immutable audit log	On each auth use case completion
`${KAFKA_TOPIC_NOTIFICATION_SEND}`	Notification dispatch requests	On OTP send and account lifecycle events

Kafka adapter: `AuditEventKafkaAdapter` (implements `AuditEventPublisher`). Fire-and-forget — publish failures are logged at `ERROR` but never roll back the primary DB transaction.

Kafka consumed: None in Phase 1.

Database: `auth_schema` (PostgreSQL)

TABLE	KEY COLUMNS
`auth_schema.users`	`id UUID PK`, `email TEXT UNIQUE`, `password_hash TEXT` (Argon2id), `role TEXT`, `center_id UUID`, `status TEXT`, `version BIGINT` (`@Version`), `deleted_at TIMESTAMPTZ`

Flyway migrations path: `classpath:db/migration/auth`

Cache (Redis):

USAGE	PORT INTERFACE	ADAPTER CLASS	KEY PATTERN	TTL
Refresh token store	`TokenStore`	`RefreshTokenRedisAdapter`	`refresh:{tokenId}`	`${JWT_REFRESH_TOKEN_EXPIRY_SECONDS}`
OTP value	`OtpStore`	`OtpRedisAdapter`	`otp:{purpose}:{userId}`	`${OTP_EXPIRY_SECONDS}`
OTP attempt counter	`OtpStore.incrementAttempts()`	`OtpRedisAdapter`	`otp:attempts:{purpose}:{userId}`	Aligned with OTP TTL

Outbound ports → adapters:

DOMAIN PORT INTERFACE	INFRASTRUCTURE ADAPTER
`UserRepository`	`UserPersistenceAdapter` → package-private `SpringDataUserRepository`
`PasswordHasher`	`Argon2PasswordHasherAdapter` (Argon2id, all parameters externalized)
`AccessTokenGenerator`	`JwtTokenProvider` (JJWT 0.12.x RS256 signing with `${JWT_PRIVATE_KEY_PATH}`)
`TokenStore`	`RefreshTokenRedisAdapter`
`OtpStore`	`OtpRedisAdapter`
`AuditEventPublisher`	`AuditEventKafkaAdapter`
`CaptchaVerifier`	`HCaptchaRestAdapter`
`NotificationSender`	`SmtpNotificationAdapter`

2.3 center-svc

Purpose: Operational backbone. Owns all physical and academic infrastructure: coaching centres, teacher rosters, batch schedules, fee structures, attendance records, and study content items.

Inbound:
- `api-gateway` (proxies `/api/v1/centers/**`)

Outbound HTTP (synchronous):

TARGET	CONFIG KEY	USE	CIRCUIT BREAKER
`ai-gateway-svc`	`${AI_GATEWAY_BASE_URL}`	AI completion requests for content assistance and curriculum suggestions	`ai-gateway` (Resilience4j)

Outbound async (Kafka topics published):

TOPIC (ENV VAR)	EVENTS	TRIGGER
`${KAFKA_TOPIC_CENTER_EVENTS}`	`CenterCreatedEvent`, `BatchStatusChangedEvent`, `TeacherAssignedEvent`, `ScheduleChangedEvent`, `ContentUploadedEvent`	On each center/batch/content domain operation
`${KAFKA_TOPIC_AUDIT_IMMUTABLE}`	All center events duplicated to audit	On all domain operations

Kafka consumed: None in Phase 1.

Database: `center_schema` (PostgreSQL)

TABLE	PURPOSE
`center_schema.centers`	Center master data
`center_schema.teachers`	Teacher roster per center
`center_schema.batches`	Academic batch definitions and lifecycle state
`center_schema.schedules`	Batch session schedules
`center_schema.fee_structures`	Fee plans per batch/center
`center_schema.attendance_records`	Per-student attendance per session
`center_schema.content_items`	Study material and content uploads

Flyway migrations path: `classpath:db/migration/center`

Cache (Redis): Redis connection is configured in `application.yml` but no center-specific caching is active in Phase 1.

Outbound ports → adapters:

DOMAIN PORT INTERFACE	INFRASTRUCTURE ADAPTER
`CenterRepository`	`CenterPersistenceAdapter`
`BatchRepository`	`BatchPersistenceAdapter`
`TeacherRepository`	`TeacherPersistenceAdapter`
`ScheduleRepository`	`SchedulePersistenceAdapter`
`FeeStructureRepository`	`FeeStructurePersistenceAdapter`
`AttendanceRepository`	`AttendancePersistenceAdapter`
`ContentRepository`	`ContentPersistenceAdapter`
`CenterEventPublisher`	Kafka adapter → `center-events` + `audit-immutable`

2.4 parent-svc

Purpose: Parent-facing bounded context. Manages parent profiles, student-to-parent links, fee payment records, notification preferences, and a "Parent Copilot" AI assistant feature.

Inbound:
- `api-gateway` (proxies `/api/v1/parents/**`)

Outbound HTTP (synchronous):

TARGET	CONFIG KEY	USE	CIRCUIT BREAKER
`ai-gateway-svc`	`${AI_GATEWAY_BASE_URL}`	Parent Copilot AI completion requests (progress summaries, recommendations)	`ai-gateway` (Resilience4j)

Outbound async (Kafka topics published):

TOPIC (ENV VAR)	EVENTS	TRIGGER
`${KAFKA_TOPIC_PARENT_EVENTS}`	`StudentLinkedEvent`, `LinkRevokedEvent`, `FeePaymentRecordedEvent`	On parent domain operations
`${KAFKA_TOPIC_AUDIT_IMMUTABLE}`	All parent events duplicated to audit	On all domain operations

Adapter: `ParentEventKafkaAdapter` (implements `ParentEventPublisher`). Fire-and-forget, publishes to `topicProperties.parentEvents()`.

Kafka consumed:

TOPIC (ENV VAR)	CONSUMER CLASS	PHASE 1 BEHAVIOUR	PHASE 2 INTENT
`${KAFKA_TOPIC_CENTER_EVENTS}`	`CenterEventConsumer`	Logs raw event payload (`log.info`)	Trigger push notifications to parents on `BatchStatusChangedEvent`, `ContentUploadedEvent`

Consumer group: `${PARENT_SVC_KAFKA_CONSUMER_GROUP}`, `auto-offset-reset: earliest`

Database: `parent_schema` (PostgreSQL)

TABLE	PURPOSE
`parent_schema.parent_profiles`	Parent identity and profile data
`parent_schema.student_links`	Parent → student UUID cross-references
`parent_schema.fee_payments`	Fee payment records and status
`parent_schema.notification_preferences`	Per-parent notification channel settings

Flyway migrations path: `classpath:db/migration/parent`

Cache (Redis): Connected but no parent-specific caching in Phase 1.

External (non-service):
- Firebase FCM (`${FIREBASE_PROJECT_ID}`, `${FIREBASE_SERVICE_ACCOUNT_KEY_PATH}`): mobile push notifications to parent app. Used exclusively by `parent-svc`.

Outbound ports → adapters:

DOMAIN PORT INTERFACE	INFRASTRUCTURE ADAPTER
`ParentProfileRepository`	`ParentProfilePersistenceAdapter`
`StudentLinkRepository`	`StudentLinkPersistenceAdapter`
`FeePaymentRepository`	`FeePaymentPersistenceAdapter`
`NotificationPreferenceRepository`	`NotificationPreferencePersistenceAdapter`
`ParentEventPublisher`	`ParentEventKafkaAdapter` → `parent-events` + `audit-immutable`

2.5 assess-svc

Purpose: Assessment and grading bounded context. Owns exams, question banks with vector embeddings, student enrolments, exam submission lifecycle, AI-assisted grading, and Computer Adaptive Testing (CAT) using Item Response Theory. Exposes a STOMP WebSocket relay for real-time exam sessions.

Inbound:
- `api-gateway` (proxies `/api/v1/assessments/**`)
- STOMP WebSocket connections from student clients during live exam sessions

Outbound HTTP (synchronous):

TARGET	CONFIG KEY	USE	CIRCUIT BREAKER
`ai-gateway-svc`	`${AI_GATEWAY_BASE_URL}`	AI-assisted grading of open-ended answers, question generation, embedding requests	`ai-gateway` (Resilience4j)

Outbound async (Kafka topics published):

TOPIC (ENV VAR)	EVENTS	TRIGGER
`${KAFKA_TOPIC_ASSESS_EVENTS}`	`ExamPublishedEvent`, `ExamSubmittedEvent`, `GradeIssuedEvent`	On exam/submission/grade domain operations
`${KAFKA_TOPIC_AUDIT_IMMUTABLE}`	All assess events duplicated to audit	On all domain operations

Adapter: `AssessEventKafkaAdapter` (implements `AssessEventPublisher`). Fire-and-forget, publishes to `topicProperties.assessEvents()`.

Kafka consumed:

TOPIC (ENV VAR)	CONSUMER CLASS	PHASE 1 BEHAVIOUR	PHASE 2 INTENT
`${KAFKA_TOPIC_CENTER_EVENTS}`	`CenterEventConsumer`	Logs raw event payload	Invalidate in-progress submissions for batches that close (`BatchStatusChangedEvent`)

Consumer group: `${ASSESS_SVC_KAFKA_CONSUMER_GROUP}`, `auto-offset-reset: earliest`

Database: `assess_schema` (PostgreSQL)

TABLE	PURPOSE
`assess_schema.exams`	Exam definitions (title, type, batch association, schedule)
`assess_schema.questions`	Question bank — includes `embedding_json TEXT` for similarity search (Phase 2: `vector(1536)`)
`assess_schema.exam_enrollments`	Student exam enrolment records
`assess_schema.submissions`	Exam session and submission lifecycle state machine
`assess_schema.submission_answers`	Per-question student answers
`assess_schema.grades`	Computed grades per submission

Flyway migrations path: `classpath:db/migration/assess` (V1–V7, 7 migration scripts)

pgvector embedding dimensions: `${AI_EMBEDDING_DIMENSIONS}`

CAT engine configuration (all externalized): `${CAT_MIN_QUESTIONS}`, `${CAT_MAX_QUESTIONS}`, `${CAT_INITIAL_THETA}`, `${CAT_CONVERGENCE_THRESHOLD}`

Cache (Redis): Connected but no assess-specific caching in Phase 1.

Outbound ports → adapters:

DOMAIN PORT INTERFACE	INFRASTRUCTURE ADAPTER
`ExamRepository`	`ExamPersistenceAdapter`
`QuestionRepository`	`QuestionPersistenceAdapter`
`ExamEnrollmentRepository`	`ExamEnrollmentPersistenceAdapter`
`SubmissionRepository`	`SubmissionPersistenceAdapter`
`SubmissionAnswerRepository`	`SubmissionAnswerPersistenceAdapter`
`GradeRepository`	`GradePersistenceAdapter`
`AssessEventPublisher`	`AssessEventKafkaAdapter` → `assess-events` + `audit-immutable`

2.6 psych-svc

Purpose: Psychometric profiling bounded context. Owns student psychometric profiles (Big Five: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism + RIASEC career type code), assessment sessions, and career mapping recommendations generated by the Python AI sidecar.

Inbound:
- `api-gateway` (proxies `/api/v1/psych/**`)

Outbound HTTP (synchronous):

TARGET	ADAPTER CLASS	CALL	CIRCUIT BREAKER	FAILURE MODE
`psych-ai-svc` (Python FastAPI sidecar)	`PsychAiSvcWebClientAdapter` (WebClient + `.block()`)	`POST ${PSYCH_AI_SVC_BASE_URL}/api/v1/predict-careers` with Big Five scores + RIASEC code	`psych-ai-svc` (Resilience4j)	CB open → `RuntimeException` thrown → `CareerMappingService` catches → `CareerMapping` saved as `FAILED` → HTTP 201 returned
`ai-gateway-svc`	WebClient	`${AI_GATEWAY_BASE_URL}` — AI completion requests (Phase 2: session transcript analysis)	`ai-gateway` (Resilience4j)	`AiProviderException` → HTTP 502

Request to psych-ai-svc:

```
profileId: String (UUID)
openness, conscientiousness, extraversion, agreeableness, neuroticism: double (0.0-1.0)
riasecCode: String (e.g. "RIA")
```

Response from psych-ai-svc:

```
topCareers: List<String>
reasoning: String
modelVersion: String
```

Outbound async (Kafka topics published):

TOPIC (ENV VAR)	EVENTS	KEY	TRIGGER
`${KAFKA_TOPIC_PSYCH_EVENTS}`	`PsychProfileCreatedEvent`, `SessionCompletedEvent`, `CareerMappingGeneratedEvent`	entity UUID	On each corresponding domain operation
`${KAFKA_TOPIC_AUDIT_IMMUTABLE}`	All psych events duplicated to audit	entity UUID	On all domain operations

Adapter: `PsychEventKafkaAdapter` (implements `PsychEventPublisher`). Dispatches each event type to both topics by key. Exceptions caught and logged (best-effort).

Kafka consumed:

TOPIC (ENV VAR)	CONSUMER CLASS	PHASE 1 BEHAVIOUR	PHASE 2 INTENT
`${KAFKA_TOPIC_ASSESS_EVENTS}`	`AssessEventConsumer`	Logs raw event payload	Trigger a `TRIGGERED` type session when grade patterns indicate academic stress

Consumer group: `${PSYCH_SVC_KAFKA_CONSUMER_GROUP}`, `auto-offset-reset: earliest`

Database: `psych_schema` (PostgreSQL)

TABLE	PURPOSE
`psych_schema.trait_dimensions`	Reference data — 11 seeded rows (5 Big Five + 6 RIASEC). No soft delete. No `@Version`.
`psych_schema.psych_profiles`	Core aggregate — Big Five scores, RIASEC code, `embedding_json TEXT` (Phase 2: pgvector), `UNIQUE(student_id, center_id)`
`psych_schema.session_histories`	Assessment session lifecycle: SCHEDULED → IN_PROGRESS → COMPLETED / CANCELLED
`psych_schema.career_mappings`	AI career predictions: PENDING → GENERATED / FAILED

Flyway migrations path: `classpath:db/migration/psych` (V1–V5; V2 inserts 11 reference rows into `trait_dimensions`)

Cache (Redis): Redis connection is configured but no psych-specific caching is active in Phase 1.

Outbound ports → adapters:

DOMAIN PORT INTERFACE	INFRASTRUCTURE ADAPTER
`PsychProfileRepository`	`PsychProfilePersistenceAdapter`
`TraitDimensionRepository`	`TraitDimensionPersistenceAdapter`
`SessionHistoryRepository`	`SessionHistoryPersistenceAdapter`
`CareerMappingRepository`	`CareerMappingPersistenceAdapter`
`PsychEventPublisher`	`PsychEventKafkaAdapter`
`PsychAiSvcClient`	`PsychAiSvcWebClientAdapter`

2.7 ai-gateway-svc

Purpose: Fully reactive AI gateway (Spring WebFlux). Routes completion, embedding, and career prediction requests to external LLM providers or the Python sidecar. Enforces per-requester, per-model rate limiting via Redis. Publishes AI usage audit events to Kafka.

Inbound:
- `api-gateway` (proxies `/api/v1/ai/**`)
- Direct calls from `center-svc`, `parent-svc`, `assess-svc`, `psych-svc` via `${AI_GATEWAY_BASE_URL}`

Outbound HTTP (all reactive `Mono<T>`):

TARGET	ADAPTER CLASS	ENDPOINT	AUTH	CIRCUIT BREAKER
Anthropic Claude API	`AnthropicWebClientAdapter`	`POST ${ANTHROPIC_BASE_URL}/v1/messages`	`x-api-key: ${ANTHROPIC_API_KEY}`, `anthropic-version: 2023-06-01`	`anthropic` CB + `R4J_TL_AI_TIMEOUT` time limiter
OpenAI API	`OpenAiEmbeddingAdapter`	`POST https://api.openai.com/v1/embeddings`	`Authorization: Bearer ${OPENAI_API_KEY}`	None (Phase 2)
Ollama	(future adapter)	`${OLLAMA_BASE_URL}`	None	Phase 2 placeholder
`psych-ai-svc` sidecar	`PsychAiSidecarWebClientAdapter`	`POST ${PSYCH_AI_SVC_BASE_URL}/api/v1/predict-careers`	None (internal network trust)	`psych-ai-svc` CB

The `AnthropicWebClientAdapter` raises `AiProviderException` for `LlmProvider.OPENAI` and `LlmProvider.OLLAMA` — those providers require their own adapter implementations.

Rate limiting:

- `RateLimitPort` → `RedisRateLimitAdapter` (`ReactiveRedisTemplate`).
- Per key: `{requesterId}:{modelType}` — Redis `INCR` + `EXPIRE` per sliding window.
- On Redis outage: `onErrorReturn(true)` — rate limiting **fails open** (requests allowed through). By design.

Outbound async (Kafka topics published):

TOPIC (ENV VAR)	EVENTS	KEY	TRIGGER
`${KAFKA_TOPIC_AI_GATEWAY_EVENTS}`	`AiRequestRoutedEvent` (success), `AiRequestFailedEvent` (failure)	`requestId` (UUID string)	On each AI request completion or failure
`${KAFKA_TOPIC_AUDIT_IMMUTABLE}`	Same events duplicated to audit	`requestId`	On each AI request

Adapter: `AiGatewayKafkaAdapter` (implements `AiGatewayEventPublisher`). Void (not reactive) — fire-and-forget. Exceptions caught in `sendBestEffort()` and logged, never propagated.

Kafka consumed: None in Phase 1.

Database: None. Fully stateless — no JPA, no datasource, no Flyway.

Cache (Redis):

USAGE	PORT INTERFACE	ADAPTER CLASS	KEY PATTERN
Per-requester per-model rate counter	`RateLimitPort`	`RedisRateLimitAdapter`	`rate:{requesterId}:{modelType}` — INCR + EXPIRE

API endpoints:

METHOD	PATH	USE CASE INTERFACE	PROVIDER
POST	`/api/v1/ai/completions`	`RouteCompletionUseCase`	Anthropic Claude (primary)
POST	`/api/v1/ai/embeddings`	`RouteEmbeddingUseCase`	OpenAI text-embedding
POST	`/api/v1/ai/career-predictions`	`RouteCareerPredictionUseCase`	psych-ai-svc sidecar

Outbound ports → adapters:

DOMAIN PORT INTERFACE	INFRASTRUCTURE ADAPTER
`LlmClient`	`AnthropicWebClientAdapter`
`EmbeddingClient`	`OpenAiEmbeddingAdapter`
`PsychAiSidecarClient`	`PsychAiSidecarWebClientAdapter`
`RateLimitPort`	`RedisRateLimitAdapter`
`AiGatewayEventPublisher`	`AiGatewayKafkaAdapter`

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3. Startup Order

3.1 Infrastructure Services (must reach healthy state before any application service starts)

```
Step 1 — PostgreSQL 16 (timescale/timescaledb-ha:pg16-latest)
   Health check: pg_isready -U ${POSTGRES_ROOT_USER}
   Interval: 10s | Timeout: 5s | Retries: 5 | Start period: 20s
   Extensions provisioned at first start: uuid-ossp, pgcrypto, pgvector (via init scripts)

Step 2 — Redis 7 (redis:7-alpine)
   Health check: redis-cli -a ${REDIS_PASSWORD} ping
   Interval: 10s | Timeout: 3s | Retries: 5
   Config: appendonly yes, appendfsync everysec

Step 3 — Kafka 3.6 (bitnami/kafka:3.6, KRaft mode — no ZooKeeper)
   Health check: kafka-topics.sh --bootstrap-server localhost:9092 --list
   Interval: 30s | Timeout: 10s | Retries: 5 | Start period: 30s
```

3.2 Application Services — Recommended Order

```
Step 4 — auth-svc
   Hard requires: PostgreSQL (auth_schema Flyway migrations), Redis, Kafka
   Flyway: classpath:db/migration/auth — creates auth_schema tables
   No dependency on any other application service
   Must be healthy before client sessions can be established, but other
   services do not call auth-svc at runtime (JWT validated locally)

Step 5 — center-svc
   Hard requires: PostgreSQL (center_schema Flyway migrations), Kafka
   Soft requires: ai-gateway-svc (circuit breaker absorbs absence at startup)
   Flyway: classpath:db/migration/center

Step 6 — ai-gateway-svc
   Hard requires: Redis (rate limiting), Kafka
   Soft requires: Anthropic API key valid, OpenAI API key valid, psych-ai-svc reachable
   No PostgreSQL dependency — stateless
   Should be healthy before services that call it (center-svc, parent-svc,
   assess-svc, psych-svc) handle real AI-dependent requests

Step 7 — parent-svc
   Hard requires: PostgreSQL (parent_schema Flyway), Kafka, Firebase credentials
   Soft requires: ai-gateway-svc (circuit breaker)
   Flyway: classpath:db/migration/parent
   Consumes center-events (Kafka must be healthy; auto-offset-reset: earliest)

Step 8 — assess-svc
   Hard requires: PostgreSQL (assess_schema Flyway), Kafka
   Soft requires: ai-gateway-svc (circuit breaker)
   Flyway: classpath:db/migration/assess (V1–V7)
   Consumes center-events

Step 9 — psych-ai-svc (Python FastAPI sidecar — not in this repo)
   Hard requires: ML model weights loaded
   psych-svc circuit breaker handles sidecar absence gracefully at runtime
   but sidecar should be started before psych-svc to avoid initial CB opens

Step 10 — psych-svc
   Hard requires: PostgreSQL (psych_schema Flyway), Kafka
   Soft requires: psych-ai-svc (circuit breaker absorbs absence; career
   mappings are saved as FAILED status instead of failing the request)
   Soft requires: ai-gateway-svc (circuit breaker)
   Flyway: classpath:db/migration/psych (V1–V5, V2 seeds trait_dimensions)
   Consumes assess-events

Step 11 — api-gateway
   Hard requires: Redis (rate limiting)
   Soft requires: all downstream services reachable (gateway returns 503
   per route for any unresponsive downstream)
   No PostgreSQL — stateless
   Start last — it is the public ingress point
```

3.3 Docker Compose (Local Dev)

```
# Start infrastructure and wait for health
docker compose --env-file ../../.env up -d postgres redis kafka

# Wait for health checks (postgres ~20s, kafka ~30s)
docker compose --env-file ../../.env up -d auth-svc center-svc ai-gateway-svc

# Wait for Flyway migrations to complete (check: docker compose logs auth-svc)
docker compose --env-file ../../.env up -d parent-svc assess-svc

# Start Python sidecar (separate process or container)
# docker run ... psych-ai-svc:latest

docker compose --env-file ../../.env up -d psych-svc

# Start edge gateway last
docker compose --env-file ../../.env up -d api-gateway
```

3.4 Kubernetes (Production)

- Each service pod has a `readinessProbe` on `GET /actuator/health` (HTTP 200 = ready).
- `startupProbe` with `failureThreshold: 30`, `periodSeconds: 10` provides a 5-minute window for Flyway migrations and cold DB connection establishment.
- PostgreSQL, Redis, and Kafka are deployed as separate StatefulSet/Helm-managed clusters before application services.
- `psych-ai-svc` is deployed as a sidecar container in the `psych-svc` Pod (shared network namespace, no service discovery needed) or as a separate Deployment with a ClusterIP Service.

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4. Failure Impact Analysis

4.1 PostgreSQL Down

AFFECTED SERVICE	USER-VISIBLE IMPACT	INTERNAL BEHAVIOUR
auth-svc	Cannot register or log in; cannot issue new tokens	`DataAccessException` propagates → `GlobalExceptionHandler` → HTTP 503
center-svc	All center/batch/content operations fail	HTTP 503
parent-svc	All parent profile operations fail	HTTP 503
assess-svc	All exam/submission/grade operations fail	HTTP 503
psych-svc	All profile/session/mapping operations fail	HTTP 503
api-gateway	Unaffected — no database	Routing continues; downstream services return 503
ai-gateway-svc	Unaffected — no database	AI routing continues

Note on existing JWT tokens: Tokens already issued before the outage remain cryptographically valid until their TTL expires. Downstream services can still validate them using their locally loaded public key, so read-only operations that do not require DB access may partially succeed (unlikely in practice — all services are DB-bound for their primary operations).

Recovery: HikariCP reconnects automatically when PostgreSQL recovers. Flyway does not re-run migrations — they are tracked in `flyway_schema_history`.

4.2 Redis Down

SERVICE	SUBSYSTEM AFFECTED	IMPACT	DESIGNED BEHAVIOUR
api-gateway	Token-bucket rate limiting	Rate limiting **fails open**	Requests pass through unmetered. Documented design: no traffic loss over rate-limit enforcement
auth-svc	Refresh token store	`TokenStore.find()` fails → token refresh fails; `TokenStore.deleteAllForUser()` fails → logout fails	HTTP 500 on refresh/logout endpoints
auth-svc	OTP store	`OtpStore.save()` and `OtpStore.find()` fail → OTP flow fails entirely	HTTP 500 on OTP endpoints; registration, password reset, MFA login blocked
ai-gateway-svc	Per-requester rate limit counters	Rate limiting **fails open** (`onErrorReturn(true)` in reactive chain)	All AI requests allowed through regardless of quota
center-svc	None active Phase 1	No impact	
parent-svc	None active Phase 1	No impact	
assess-svc	None active Phase 1	No impact	
psych-svc	None active Phase 1	No impact	

Recovery: Redis reconnects automatically. Rate-limit counters reset on restart (sliding window starts fresh — acceptable given eventual-consistency rate-limiting design). Refresh tokens stored in Redis are lost on Redis restart if AOF is not enabled or the node is cold-started; users will need to re-authenticate.

4.3 Kafka Down

IMPACT CATEGORY	DETAIL
Domain operations	All user-facing operations succeed. The `@Transactional` DB commit completes before Kafka publish is attempted.
Audit events	All audit events are silently dropped for the duration of the outage. Logged at `ERROR` level. No compensation mechanism in Phase 1 (outbox pattern is a Phase 2 upgrade path).

IMPACT CATEGORY	DETAIL
parent-svc `CenterEventConsumer`	Stops receiving center-events. Phase 1: no user impact. Phase 2: parent push notifications stop firing.
assess-svc `CenterEventConsumer`	Stops receiving center-events. Phase 1: no user impact. Phase 2: batch-close invalidation of submissions stops.
psych-svc `AssessEventConsumer`	Stops receiving assess-events. Phase 1: no user impact. Phase 2: grade-triggered sessions stop firing.

Recovery: Kafka consumers resume from their last committed offset (`auto-offset-reset: earliest` for new groups ensures no messages from before first launch are skipped). Events published while Kafka was down are permanently lost — there is no at-least-once guarantee in Phase 1.

4.4 psych-ai-svc (Python Sidecar) Down

CALLER	IMPACT	BEHAVIOUR
psych-svc	Career mapping requests fail gracefully	`PsychAiSvcWebClientAdapter.predictCareers()` is annotated `@CircuitBreaker(name = "psych-ai-svc")`. When the CB opens, it throws `RuntimeException`. `CareerMappingService.requestCareerMapping()` catches it → calls `mapping.fail()` → saves `CareerMapping` with status `FAILED` → DB transaction commits → returns HTTP 201 with `FAILED` status in response body. No HTTP 5xx to the client.
ai-gateway-svc	Career prediction routing fails	`PsychAiSidecarWebClientAdapter.predictCareers()` → CB opens → `AiProviderException` → HTTP 502 returned to caller.

Recovery: Both circuit breakers transition to HALF_OPEN after `wait-duration-in-open-state`. First successful probe closes the circuit. `FAILED` career mappings must be explicitly re-requested by the student — there is no automatic retry or background job in Phase 1.

4.5 Anthropic API Down

AFFECTED PATH	IMPACT	BEHAVIOUR
ai-gateway-svc `/api/v1/ai/completions`	LLM completions fail	`AnthropicWebClientAdapter` receives error → `AiProviderException("Anthropic API error [status]: body")` → circuit breaker `anthropic` increments failure count → on threshold: CB opens → all subsequent completion requests fail fast with `CallNotPermittedException` → `AiProviderException` → HTTP 502.
center-svc, parent-svc, assess-svc calling ai-gateway-svc	AI-dependent features degrade	Each service has its own `ai-gateway` circuit breaker. On enough 502s from ai-gateway-svc, each service's CB opens, failing fast. Core non-AI operations (CRUD) continue to succeed.
Ollama fallback	Not automatic in Phase 1	`LlmProvider.OLLAMA` raises `AiProviderException("Ollama routing not implemented")` — there is no automatic fallback from Anthropic to Ollama. Phase 2 upgrade: implement `OllamaWebClientAdapter` and add provider selection logic.

4.6 OpenAI API Down

AFFECTED PATH	IMPACT	BEHAVIOUR
ai-gateway-svc `/api/v1/ai/embeddings`	Embedding requests fail	`OpenAiEmbeddingAdapter` returns error → `AiProviderException` → HTTP 502. No circuit breaker on OpenAI in Phase 1.
assess-svc question similarity search	Embedding indexing fails	New question embeddings cannot be generated; similarity search returns degraded results. Existing embeddings are unaffected.
psych-svc embedding enrichment	Embedding field remains null	`embedding_json` column stays null until OpenAI recovers. Phase 2: pgvector HNSW index will be affected.

4.7 hCaptcha API Down

AFFECTED ENDPOINTS	IMPACT	BEHAVIOUR
auth-svc `/api/v1/auth/register`, `/api/v1/auth/login`	New registrations and captcha-verified logins blocked	`HCaptchaRestAdapter` circuit breaker `captcha-client` opens → fallback method `captchaFallback()` returns `false` → `CaptchaVerifier.verify()` returns `false` → auth service rejects the request with HTTP 400 "Invalid captcha."

Design rationale for fail-closed: The hCaptcha circuit breaker deliberately fails closed (rejects requests) rather than open (allows). Allowing unverified registrations during an outage would defeat bot protection, which is a security requirement.

4.8 SMTP Server Down

AFFECTED FLOWS	IMPACT	BEHAVIOUR
auth-svc OTP email delivery	Email verification, password reset, MFA login via email OTP blocked	`SmtpNotificationAdapter.sendOtpEmail()` throws `MessagingException` → wrapped as `IllegalStateException("Email delivery failed")` → propagates up the call stack → HTTP 500. No circuit breaker on SMTP — the exception propagates directly.

SMS OTP path via Twilio is unaffected by SMTP outage.

4.9 api-gateway Down

All external client traffic is blocked. Downstream services continue operating normally — they are not aware the gateway is down. Internal service-to-service calls (psych-svc → psych-ai-svc, any service → ai-gateway-svc) are unaffected. Re-deploy the gateway pod to restore external access; downstream service state is unaffected.

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5. External Dependencies

5.1 Anthropic API (Claude LLM)

ATTRIBUTE	DETAIL
Purpose	Primary LLM for text completion requests
Caller	`ai-gateway-svc` — `AnthropicWebClientAdapter`
Base URL	`${ANTHROPIC_BASE_URL}` (e.g. `https://api.anthropic.com`)
Endpoint	`POST /v1/messages`
Auth headers	`x-api-key: ${ANTHROPIC_API_KEY}`, `anthropic-version: 2023-06-01`
Model	`${ANTHROPIC_MODEL}` (e.g. `claude-sonnet-4-6`)
Request format	`{ model, max_tokens, messages: [{role: "user", content: "[System]: {system}\n\n{user}"}] }`
Response parsing	`content[0].text` for completion text; `usage.input_tokens` + `usage.output_tokens` for cost tracking
Circuit breaker	`anthropic` (Resilience4j) + `anthropic` time limiter (`${R4J_TL_AI_TIMEOUT}`)
Error mapping	Non-2xx → `AiProviderException("Anthropic API error [status]: body")` → HTTP 502
Token tracking	Latency (ms), input tokens, output tokens included in `CompletionResponse` record and `AiRequestRoutedEvent`

5.2 OpenAI API (Embeddings)

ATTRIBUTE	DETAIL
Purpose	Vector embedding generation for question similarity search and content retrieval
Caller	`ai-gateway-svc` — `OpenAiEmbeddingAdapter`
Base URL	`https://api.openai.com` (hardcoded)
Endpoint	`POST /v1/embeddings`
Auth header	`Authorization: Bearer ${OPENAI_API_KEY}`
Model	`${AI_EMBEDDING_MODEL}` (e.g. `text-embedding-3-small`)
Dimensions	`${AI_EMBEDDING_DIMENSIONS}` (e.g. 1536) — must match the `vector(N)` column in assess_schema and psych_schema
Request format	`{ model, input: "text string" }`
Response parsing	`data[0].embedding` — `List<Double>` of dimension N
Error mapping	Non-2xx → `AiProviderException("OpenAI Embeddings API error [status]: body")` → HTTP 502
Latency tracking	`latencyMs` calculated from `startMs` to response, included in `EmbeddingResponse`

5.3 Ollama (Self-Hosted LLM)

ATTRIBUTE	DETAIL
Purpose	Local LLM for development environments and offline fallback
Caller	`ai-gateway-svc` (adapter not yet implemented in Phase 1)
Base URL	`${OLLAMA_BASE_URL}`
Auth	None (local service)
Phase 1 status	Placeholder — `LlmProvider.OLLAMA` in `AnthropicWebClientAdapter.complete()` raises `AiProviderException("Ollama routing not implemented in this adapter")`. No automatic failover from Anthropic.
Phase 2 plan	Implement `OllamaWebClientAdapter implements LlmClient` + routing logic in `LlmRoutingService` to select Ollama when `${LLM_PROVIDER}` env var = `OLLAMA` or as a fallback on Anthropic CB open.

5.4 psych-ai-svc (Python FastAPI Sidecar)

ATTRIBUTE	DETAIL
Purpose	Career prediction ML model — Big Five personality scores + RIASEC code → ranked career suggestions with AI reasoning
Technology	Python FastAPI (not in this repository)
Default port	8095
Deployment	Sidecar container alongside `psych-svc` Pod in Kubernetes, or standalone container in docker-compose
Callers	`psych-svc` (`PsychAiSvcWebClientAdapter`) and `ai-gateway-svc` (`PsychAiSidecarWebClientAdapter`)
Endpoint	`POST /api/v1/predict-careers`
Auth	None (internal network trust; mTLS is a Phase 2 hardening step)
Connect timeout	`${PSYCH_AI_SVC_CONNECT_TIMEOUT_SECONDS}` (both callers)
Read timeout	`${PSYCH_AI_SVC_READ_TIMEOUT_SECONDS}` (both callers)
Circuit breaker	`psych-ai-svc` on both `psych-svc` and `ai-gateway-svc`

Request payload:

```
{
  "profileId": "550e8400-e29b-41d4-a716-446655440000",
  "openness": 0.75,
  "conscientiousness": 0.82,
  "extraversion": 0.60,
  "agreeableness": 0.70,
  "neuroticism": 0.35,
  "riasecCode": "RIA"
}
```

Response payload:

```
{
  "topCareers": ["Software Engineer", "Data Scientist", "Research Scientist"],
  "reasoning": "High openness and conscientiousness combined with Investigative RIASEC code...",
  "modelVersion": "psych-ai-v2"
}
```

Failure modes:

- `psych-svc` caller: circuit breaker opens → `CareerMappingService` catches → `CareerMapping.fail()` → status `FAILED` saved → HTTP 201 returned with FAILED body.
- `ai-gateway-svc` caller: circuit breaker opens → `AiProviderException` → HTTP 502 returned to caller.

5.5 hCaptcha Enterprise API

ATTRIBUTE	DETAIL
Purpose	Bot protection on user registration and login endpoints

ATTRIBUTE	DETAIL
Caller	`auth-svc` — `HCaptchaRestAdapter` (Spring `RestClient`, blocking)
Endpoint	`POST ${HCAPTCHA_VERIFY_URL}` (e.g. `https://hcaptcha.com/siteverify`)
Request format	`application/x-www-form-urlencoded`: `secret=${HCAPTCHA_SECRET_KEY}`, `response={captchaToken}`, `remoteip={clientIp}`
Auth	Secret key embedded in request body
Site key	`${HCAPTCHA_SITE_KEY}` — distributed to front-end clients for widget rendering
Circuit breaker	`captcha-client` — fallback `captchaFallback()` returns `false` (fail **closed**: reject all requests during outage)
Success condition	`HCaptchaResponse.success() == true`

5.6 SMTP Server (Email OTP Delivery)

ATTRIBUTE	DETAIL
Purpose	OTP delivery for email verification, password reset, and MFA login
Caller	`auth-svc` — `SmtpNotificationAdapter` (JavaMailSender, blocking)
Config	`${MAIL_HOST}`, `${MAIL_PORT}`, `${MAIL_USERNAME}`, `${MAIL_PASSWORD}`
Protocol	SMTP with STARTTLS (`mail.smtp.auth: true`, `mail.smtp.starttls.enable: true`)
From identity	`${MAIL_FROM_ADDRESS}`, `${MAIL_FROM_NAME}`
OTP purposes mapped	`EMAIL_VERIFICATION` → "Verify your EduTech account"; `PASSWORD_RESET` → "Reset your EduTech password"; `LOGIN_MFA` → "Your EduTech login code"
Failure mode	`MessagingException` → `IllegalStateException("Email delivery failed")` → HTTP 500 (no circuit breaker)

5.7 Twilio Verify API (SMS OTP)

ATTRIBUTE	DETAIL
Purpose	SMS OTP delivery for MFA login and phone number verification
Caller	`auth-svc` — `SmtpNotificationAdapter.sendOtpSms()`
Config	`${TWILIO_ACCOUNT_SID}`, `${TWILIO_AUTH_TOKEN}`, `${TWILIO_VERIFY_SERVICE_SID}`
Production behaviour	Twilio SDK sends SMS via Verify API
Development behaviour	OTP logged to console: `log.info("[DEV-ONLY] SMS OTP for to={} purpose={} otp={}", ...)` — no actual SMS sent

5.8 Firebase Cloud Messaging (Push Notifications)

ATTRIBUTE	DETAIL
Purpose	Mobile push notifications to the parent mobile app (Android / iOS)
Caller	`parent-svc` exclusively (Firebase Admin SDK)
Config	`${FIREBASE_PROJECT_ID}`, `${FIREBASE_SERVICE_ACCOUNT_KEY_PATH}`
Usage	Parent Copilot notifications, batch status change alerts (Phase 2 integration with `CenterEventConsumer`)
Note	Only `parent-svc` has Firebase credentials configured; no other service uses Firebase

See also: `/docs/ARCHITECTURE.md` for the full system architecture document and `docs/services/*/COMPLETION.md` for per-service frozen implementation records.